

Smart Air Monitoring and Dust Filtration System for Industrial Workers

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Abstract - Industrial workers operating in dust-intensive environments such as textile mills, rice mills, cement plants, and match factories are continuously exposed to hazardous airborne particulate matter, leading to severe occupational health issues. Conventional industrial air purification systems are often expensive, bulky, and unsuitable for small- and medium-scale industries. This paper proposes a low-cost, microcontroller-based Smart Air Monitoring and Dust Filtration System designed to continuously monitor air quality and automatically activate air filtration when pollution levels exceed predefined safety thresholds. The system employs a particulate matter (PM2.5) dust sensor, gas sensor, temperature and humidity sensor, and an Arduino Uno microcontroller to process real-time environmental data. A relay-controlled exhaust fan and filtration unit are activated during hazardous conditions, while an LCD display and buzzer provide immediate alerts to workers. Experimental results demonstrate reliable dust detection, fast response time, and effective air quality improvement, proving the suitability of the proposed system for enhancing worker safety in industrial environments.

Index Terms - Air Quality Monitoring, PM2.5, Industrial Safety, Dust Filtration, Arduino Uno, Embedded Systems, IoT

INTRODUCTION

Air pollution in industrial workplaces is one of the most critical occupational health challenges faced by workers today. Industries such as textile manufacturing, rice milling, cement processing, and metal fabrication generate large quantities of fine dust, smoke, and toxic gases during daily operations. Continuous exposure to particulate matter (PM2.5 and PM10) can result in respiratory diseases such as asthma, chronic bronchitis, silicosis, and long-term cardiovascular complications. Most existing industrial air purification solutions are designed for large-

scale facilities and require high installation and maintenance costs. As a result, many small- and medium-scale industries lack effective air quality monitoring and filtration systems, leaving workers exposed to unsafe air conditions. Moreover, manual monitoring methods fail to provide real-time detection and timely response to sudden pollution spikes. To overcome these limitations, this paper presents a Smart Air Monitoring and Dust Filtration System that combines real-time sensing, automated control, and alert mechanisms into a compact and cost-effective solution. The proposed system continuously monitors air quality parameters and activates filtration only when required, ensuring both worker safety and energy efficiency. IoT-based air quality monitoring systems integrated with AI, cloud, and blockchain enable real-time, scalable, and secure pollution monitoring and management, despite challenges in sensor accuracy, privacy, and energy efficiency [1]. A fog-enabled IoT air quality monitoring and forecasting system using a BiLSTM deep learning model provides low-latency real-time monitoring and accurate AQI prediction, outperforming baseline models and enabling timely pollution control decisions [2]. An IoT- and machine learning-based air quality monitoring system using MQ sensors, WSN with LoRa communication, cloud data sharing, and recurrent neural networks can enable real-time pollution prediction and smart traffic management to effectively mitigate urban air pollution [3]. A low-cost, energy-efficient IoT-based air quality monitoring system is proposed to overcome the limitations of conventional methods by enabling real-time, high-resolution air quality data collection suitable for smart city applications [4]. A portable, low-power air monitoring and purification system is proposed that reduces air pollution by passing contaminated air through multiple filtration stages based on detected pollution concentration levels [5]. This study demonstrates that low-cost electrochemical gas sensors in WSN-based air quality monitoring are affected by aging and proposes adaptive calibration and compensation formulae for NO₂ and O₃ sensors to improve long-term measurement accuracy and reliability [6]. This paper reviews next-generation air pollution monitoring systems using MEMS and WSN

technologies, classifies them into static, community, and vehicle sensor networks, and highlights their limitations and future research objectives for achieving high-resolution, flexible air quality monitoring [7]. An IoT-based air quality monitoring system using gas sensors, GPS, and an ESP8266 module enables low-cost, location-aware detection of atmospheric pollution levels with cloud connectivity, supporting smart city initiatives for pollution control [8]. This paper presents a low-cost urban air pollution monitoring and forecasting system that uses wireless sensor motes and machine learning models, showing that multivariate forecasting with the M5P algorithm achieves the best prediction accuracy for pollutants such as O_3 , NO_2 , and SO_2 [9]. This paper proposes an IoT-based industrial air quality monitoring system with compact sensing devices and machine learning analytics, demonstrating that Gaussian Process Regression achieves the highest accuracy for predicting pollutant levels and potential safety threshold violations [10]. An automated water tank cleaning system using digital twin technology and IoT enables real-time monitoring, predictive maintenance, optimized cleaning, reduced manual effort, and sustainable water quality management [11].

PROPOSED SYSTEM ARCHITECTURE

The proposed Smart Air Monitoring and Dust Filtration System is designed to provide continuous air quality monitoring and automatic dust filtration in industrial environments. The system architecture consists of four major components: sensing unit, control unit, actuation unit, and alert/display unit in fig.1.

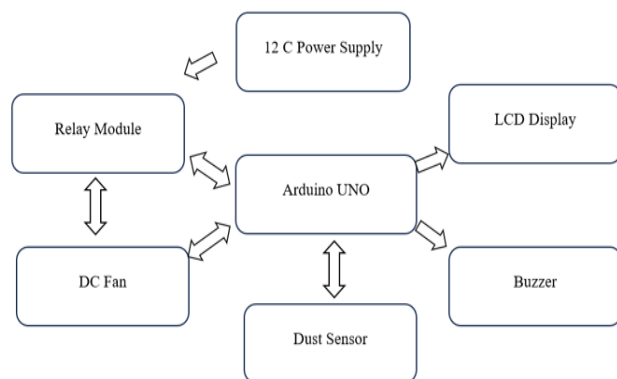


Fig 1. Flow Chart

The sensing unit includes a PM2.5 dust sensor for measuring fine particulate matter, a gas sensor for detecting harmful gases, and a temperature and humidity sensor for environmental monitoring. These sensors generate analog signals corresponding to pollutant concentration levels. The control unit is based on an Arduino Uno microcontroller, which processes sensor data in real time. When the measured dust concentration exceeds predefined threshold values, the microcontroller activates the filtration mechanism. The actuation unit consists of a relay module and a 12V DC

exhaust fan coupled with an air filter. The fan draws polluted air through the filtration unit, reducing dust concentration in the surrounding environment. The alert and display unit includes an I²C LCD display for real-time data visualization and a buzzer for immediate warning during hazardous conditions.

HARDWARE AND METHODOLOGY

The system operates using a regulated power supply, providing 5V for sensors and control units and 12V for the exhaust fan. Upon powering the system, the microcontroller initializes all sensors and performs a basic self-test to ensure proper operation.

A. Required Components

Arduino Uno: It is an open-source microcontroller development board based on the ATmega328P microcontroller, widely used for embedded system prototyping and research applications due to its simplicity, low cost, and flexibility. It operates at a clock frequency of 16 MHz with an operating voltage of 5 V and provides 32 KB of flash memory, 2 KB of SRAM, and 1 KB of EEPROM for program and data storage. The board features 14 digital input/output pins, of which 6 support pulse-width modulation (PWM), along with 6 analog input pins having 10-bit resolution, enabling effective interfacing with sensors and actuators. Arduino Uno supports serial communication protocols such as UART, SPI, and I²C, facilitating seamless integration with peripheral devices and communication modules shown in fig.2.



Fig 2. Arduino UNO

PM2.5 Sensor: A sensing device used to measure the concentration of fine particulate matter with an aerodynamic diameter of 2.5 micrometers or less, which poses significant health and environmental risks due to its ability to penetrate deep into the human respiratory system. These sensors typically operate based on laser scattering or optical detection principles, where airborne particles pass through a light beam and the scattered light intensity is analyzed to estimate particle concentration, usually expressed in micrograms per cubic meter ($\mu g/m^3$) shown in fig.3.



Fig 3. PM 2.5 Sensor

Single-channel Relay Module: It is an electromechanical switching device used to control high-voltage or high-current

loads using a low-power control signal from a microcontroller or embedded system. It typically consists of a relay, driver transistor, optocoupler for electrical isolation, flyback diode for protection, and indicator LEDs for status monitoring. The module allows safe interfacing between low-voltage control circuits (such as 5 V or 3.3 V logic) and AC or DC loads, commonly rated up to 250 V AC or 30 V DC depending on the relay specifications shown in fig.4.



Fig 4. Single Channel Relay Module

12 V DC Fan: It is a compact electromechanical device used for ventilation and thermal management by converting electrical energy into controlled airflow. It operates on a nominal 12-volt direct current supply and typically consists of a brushless DC motor, rotor blades, and an electronic commutation circuit to ensure efficient and reliable operation. These fans are widely used in embedded systems, power electronics, computer hardware, and industrial equipment for heat dissipation and air circulation in fig.5.



Fig 5. 12 V DC Fan

Air Filter: A simple air filter is a basic filtration device designed to remove dust, particulate matter, and other airborne impurities from the air to improve air quality. It typically consists of a porous filtering medium such as paper, foam, fabric, or fibrous material that traps particles as air passes through it while allowing clean air to flow out. Simple air filters are commonly used in ventilation systems, air purifiers, and low-cost pollution control setups due to their ease of construction, low maintenance requirements, and cost-effectiveness in fig.6.



Fig 6. Air Filter

LCD I²C Display: It is a character display module that uses the Inter-Integrated Circuit (I²C) communication two data lines, namely Serial Data (SDA) and Serial Clock (SCL). It typically consists of a standard LCD (such as 16×2 or 20×4)

integrated with an I²C backpack module based on controllers like the PCF8574, which converts parallel data into serial communication. The use of I²C significantly reduces the number of required I/O pins, simplifying circuit design and enabling easy integration with embedded systems in fig.7.



Fig 7. LCD I²C Display

Buzzer: An audio signalling device used to generate sound alerts in electronic systems for indication and warning purposes. It operates by converting electrical energy into audible sound, commonly using piezoelectric or electromagnetic principles, and is widely used in embedded systems, alarms, and monitoring applications to provide user feedback or fault notifications. Buzzers are valued for their low power consumption, compact size, and ease of interfacing with microcontrollers.

Power Supply: It converts electrical power from a source, such as AC mains or a battery, into a regulated DC voltage suitable for system components. In embedded systems, a stable power supply ensures reliable operation, protects components from voltage fluctuations, and supports efficient performance of sensors, actuators, and control units.

B. Software Required

The software environment for the IoT-based air purifier is centred around Arduino IDE and C/C++ programming, enabling seamless integration between sensors, actuators, and displays for real-time air quality management. Arduino IDE is the official development environment for programming Arduino boards like the UNO. Provides a code editor with syntax highlighting, error checking, and autocompletion. Includes a compiler and uploader to transfer sketches to the Arduino microcontroller. Serial monitor allows real-time debugging and monitoring of sensor data. Arduino uses a simplified version of C/C++, which is compiled into machine code for the microcontroller. Handles sensor data acquisition from PM2.5, MQ-2, MQ-3, and other sensors. Controls actuators such as the relay module, 12V DC fan, buzzer, and display. Implements conditional logic for threshold-based automatic control. Sensor data acquisition is performed continuously using analog inputs. The PM2.5 dust sensor measures airborne particulate concentration and outputs a voltage proportional to dust density. The Arduino converts this analog signal into digital values and compares them against predefined safety thresholds.

When the dust level remains within acceptable limits, the system operates in monitoring mode. If the dust concentration exceeds the threshold, the microcontroller triggers the relay module, activating the DC fan and filtration unit. Simultaneously, the buzzer is activated to alert workers, and real-time values are displayed on the LCD. This automated control strategy ensures rapid response to pollution spikes while minimizing unnecessary power consumption shown in fig.8.

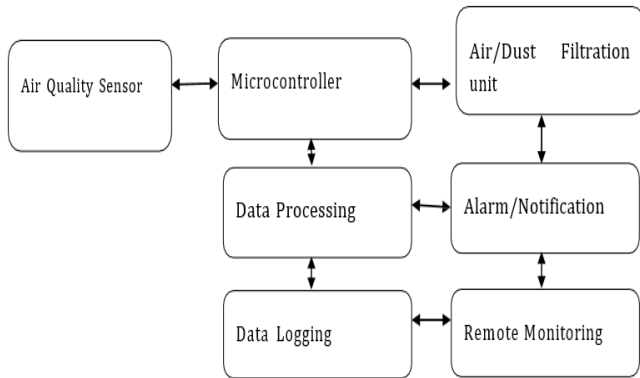


Fig 8. Methodology

EXPERIMENTAL RESULTS AND DISCUSSION

The proposed system was tested under different air quality conditions to evaluate its performance. During moderate pollution levels, the system detected dust concentrations of approximately $98 \mu\text{g}/\text{m}^3$ and operated the fan intermittently to maintain air quality. Under heavily polluted conditions, dust levels reached approximately $255 \mu\text{g}/\text{m}^3$, triggering continuous fan operation and buzzer alerts. The Smart Air Monitoring and Dust Filtration System is designed to continuously monitor air quality in industrial environments and automatically control air purification based on the pollution level. The core of the system is the Arduino Uno, which acts as the main controller and coordinates all sensor readings and actions. The GP2Y1010AU0F Dust Sensor (PM2.5/PM3.5) detects the amount of dust and fine particulate matter present in the air. It sends an analog signal to the Arduino, which converts it into digital values to determine the air quality index. When the detected dust level exceeds a pre-set threshold, the Arduino activates the 1-Channel Relay Module, which in turn switches on the 12V DC fan.

The fan helps to remove dust particles and circulate clean air, thereby improving the working environment. The air quality data is displayed in real-time on the I²C LCD Display (16x2 or 20x4), allowing workers or supervisors to easily monitor current air conditions. If the air quality becomes hazardous, the buzzer automatically sounds an alarm, alerting nearby workers to take safety precautions or vacate the area. The system continuously monitors and updates readings, ensuring accurate and reliable detection of air pollution levels. Because it is based on Arduino Uno, the system is easy to program, compact, and cost-effective, making it ideal for industrial applications.



Fig 9. Simulation

The response time of the system was observed to be between 5 and 10 seconds from pollution detection to filtration activation. The LCD display provided clear real-time visualization of dust concentration and system status. The results confirm that the system effectively detects variations in air quality and responds automatically to protect workers from harmful dust exposure in fig.9.



Fig 10. LCD display of moderate air density level

This indicates that the air quality is moderately polluted. The system detects an increased dust concentration but still within the acceptable range. At this stage, the fan operates intermittently to maintain air purity in fig.10 and fig.11.



Fig 11. LCD Display High Air Density Level

This reading represents a high dust concentration level, signifying a severely polluted environment. When such a value is detected, the Arduino immediately triggers the relay module to activate the 12V DC exhaust fan and the buzzer, providing both filtration and an alert to workers. The consistent voltage output (~4.48V) corresponds to the sensor's analog output in proportion to dust density. The increasing $\mu\text{g}/\text{m}^3$ value clearly demonstrates the sensor's responsiveness to particulate matter concentration. This validates the system's ability to detect varying air quality conditions and automatically respond to protect industrial workers shown in fig.12.

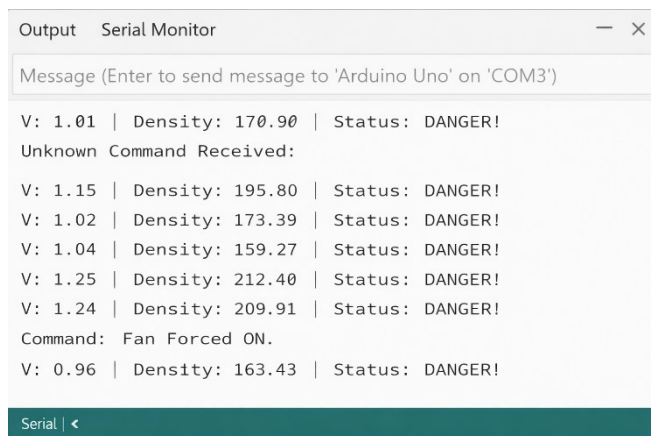


Fig 12. Serial Monitor Reading Of Air Density Level

CONCLUSION

This paper presented a Smart Air Monitoring and Dust Filtration System to improve worker safety in industrial environments through real-time air quality monitoring and automated dust control. The system integrates a GP2Y1010AU0F dust sensor, Arduino Uno, relay-controlled filtration fan, LCD display, and buzzer to detect particulate matter, trigger automatic filtration, and provide immediate alerts when pollution levels exceed safe thresholds. Experimental results demonstrated reliable performance, fast response time, stable operation, and low power consumption. Owing to its compact design, affordability, and ease of deployment, the proposed system is well suited for small- and medium-scale industries, where it effectively reduces health risks, minimizes manual supervision, and contributes to a safer and cleaner working environment. Although the Smart Air Monitoring and Dust Filtration System has successfully achieved its objectives, there is considerable scope for further enhancement and development. Future improvements can focus on increasing the system's efficiency, connectivity, and usability for large-scale industrial applications. One major enhancement could be the integration of IoT (Internet of Things) technology using modules such as ESP8266 or ESP32 to enable cloud-based monitoring and mobile app connectivity. This would allow users to access real-time air quality data remotely, receive alerts through smartphones, and analyze long-term pollution trends for preventive maintenance. Another area of improvement involves upgrading the sensor network. Advanced sensors like MQ135, CO_2 , or SO_2 gas sensors can be added to detect multiple harmful gases alongside dust particles, making the system suitable for chemical and manufacturing industries.

The filtration mechanism can also be enhanced by including multi-stage filters or electrostatic air purifiers to improve dust removal efficiency. Additionally, the system could be powered through renewable energy sources such as solar panels, promoting sustainability and reducing power consumption. The inclusion of AI and machine learning algorithms could enable predictive analysis allowing the system to anticipate air quality deterioration and take corrective actions.

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